atomQUEST 2026

AtomQuest 2026 is a product-based challenge built around three of Atomberg's own product lines - Fans, Water Purifiers, and Mixer Grinders. Every team chooses ONE track and builds a real, working prototype addressing a genuine problem within it. This is not a case study or a slide deck - the challenge ends with a functioning product, demonstrated live.

TRACK 1 · FAN

Air, Reimagined for the Next Decade.

"From Fans to Intelligent Comfort Ecosystems."

CHALLENGE STATEMENT

Atomberg's fan portfolio already spans BLDC ceiling fans with remote, app, and voice control; low-noise BLDC exhaust fans built for efficient ventilation; wide-coverage wall and pedestal fans with multi-speed remotes; and even battery-powered personal fans built for hours of cordless, silent use. As energy consumption patterns, connected homes, and consumer expectations continue to evolve, there's a growing need to push this even further. Participants are invited to explore how Atomberg can reimagine the fan category and transform it into a smarter, more integrated solution that enhances consumers' lives over the next 5–10 years.

AREAS OF EXPLORATION

- Blade design for energy efficiency, higher air throw, low noise
- Fan as an enabler for smart home - ambient temperature & humidity-based triggers to control connected devices
- Pedestal fans with multi-directional swing motion; dynamic or sensor-based control of swing span
- Sensor enabled exhaust fans with auto ON/OFF

TRACK 2 · WATER PURIFIER

Rewriting the Rules of Clean Water.

"From Purifiers to Intelligent Water Ecosystems."

CHALLENGE STATEMENT

Atomberg's water purifiers already combine RO, UF, UV, and Alkaliser stages across a 7-stage filtration system, with smart TDS-based modes, IoT connectivity, and technology built to retain minerals rather than strip them. As water quality concerns and consumer expectations continue to evolve, there's a growing need to push this even further. Participants are invited to explore how Atomberg can reimagine the water purifier category and transform it into a smarter, more integrated solution that enhances consumers' lives over the next 5–10 years.

AREAS OF EXPLORATION

- Input water quality sensing beyond TDS
- Filtered water enhancement for rejected minerals
- DIY solution for filter replacement
- Prognostics of filter failure using data analytics

TRACK 3 · MIXER GRINDER

Reimagine the Grind.

"From Mixer Grinders to Intelligent Kitchen Companions."

CHALLENGE STATEMENT

Atomberg's mixer grinders already go beyond the basics - a dedicated Coarse Mode for authentic, silbatta-like textures, a versatile speed range, and built-in safety features designed to handle every kitchen task. As Indian kitchens and cooking habits continue to evolve, there's a growing need to push this even further. Participants are invited to explore how Atomberg can reimagine the mixer grinder category and transform it into a smarter, more integrated solution that enhances consumers' lives over the next 5–10 years.

AREAS OF EXPLORATION

- Grind & Texture
- Noise Reduction
- Heat reduction in ingredients during grinding
- Improving jar reliability & durability
- Garlic peeling for Indian garlic & safer coconut grating attachments
- Safety & usability (non-contact / non-invasive lid detection)

Choose one track and one problem area within it as your starting point. You're welcome to combine ideas across problem areas within the same track if your product genuinely addresses more than one.

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STAGE 1 · ABSTRACT SUBMISSION

Deadline: 11 October 2026 | **Format:** two/three pager (excluding intro), in PDF format

File name: TeamName_TrackNo e.g. **Aerominds_Track1.pdf** | Track 1 Fan · Track 2 Water Purifier · Track 3 Mixer Grinder

The abstract must contain at least the following:

A Team & Institutional Context

- Team members' details
- Institute details

B Problem & Idea Framing

- Track and specific Area of Exploration being addressed - if combining more than one, state which and why
- Proposed idea/mechanism, in one paragraph - what you're building and how it works
- Baseline-vs-target performance claim, quantified (e.g., “reduce noise from X dB to Y dB”)
- Any other technical detail you consider relevant, beyond the points above

C Technical Feasibility

Provide a paragraph explaining **how your product works and why it is technically feasible** - the core mechanism or technology you are using, the key components and subsystems involved, and how any safety-critical aspects of the design are handled. Include **relevant numerical specifications** wherever they are necessary to support your claims.

D Execution Feasibility

- Rough list of components for your product
- System-level block diagram
- Rough cost/BOM estimate - order of magnitude, not a detailed breakdown
- Component sourcing risk - flag anything custom-fabricated, imported, or with a long lead time
- Build plan to Round 2 - how you get from this abstract to a working Level 1 prototype in a short span of time before the next deadline
- Any other technical detail you consider relevant, beyond the points above